Latent Render-and-Compare: SE(3) Diffusion with a Frozen Feature-Space World Model

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FIGURE PLACEHOLDER (teaser / method overview).
Left: an SE(3) diffusion step in LiePose [13] sees only image features and the noisy pose.
Right: LaRC at the same step. The noisy pose g_t retrieves the nearest training view $(\mathbf{z}^{\text{ref}}, g^{\text{ref}})$; the frozen world model $\mathcal{F}_\phi$ predicts the DINOv2 tokens $\hat{\mathbf{z}}(g_t)$ of the object at g_t in the observation's crop; the denoiser compares $\hat{\mathbf{z}}(g_t)$ with the observed tokens $\mathbf{z}$ and predicts the noise. Show: observation crop, retrieved reference, PCA-coloured predicted tokens at a wrong and at the converged pose, residual map $\|\hat{\mathbf{z}} - \mathbf{z}\|$.

Figure 1. **Render-and-compare inside every denoising step, in feature space.** At each step of SE(3) diffusion, a frozen world model predicts what the object would look like, as DINOv2 tokens, at the current noisy pose hypothesis; the denoiser conditions on the comparison between this prediction and the observation. No renderer, CAD texture or test-time optimisation is needed.

Abstract

Diffusion models on SE(3) handle the pose ambiguity of symmetric and occluded objects by sampling, but each denoising step is a feed-forward function of image features and the noisy pose: the network never checks whether its current hypothesis explains the image. Render-and-compare refiners do perform this check, but they require a renderer and a textured 3D model, and run as a separate stage after an initial estimate. We propose LaRC, a pose-estimation architecture built around this comparison, carried out in a learned feature space inside every denoising step. A frozen world model, trained on DINOv2 patch tokens, predicts the tokens of the object at any pose hypothesis from the nearest training view and the relative pose between them. At every denoising step and for every sample, the denoiser receives the observed tokens, the tokens predicted at its current hypothesis, and their difference. The same prediction defines an energy that ranks complete samples at test time without ground truth. The world model is trained once per dataset with a pose-contrastive objective and is never updated by the denoiser. A control that removes only the predicted tokens isolates the effect of the comparison. We evaluate on T-LESS and YCB-V under the BOP protocol, on the SPEED+ spacecraft benchmark with its official score, and on SYMSOL-T, against LiePose trained with the same budget. ***[TODO: one sentence of results, written only after the tables are filled.]***

1. Introduction

Estimating the 6D pose of a known object from a single RGB image is ambiguous whenever the visible part of the object looks the same from several viewpoints: because of symmetry, because the distinguishing features are occluded, or because a texture-less object offers little to disambiguate [9, 11, 25]. Treating the pose as a distribution rather than a single value handles this ambiguity, and diffusion models on SE(3) have become an effective way to represent such distributions. LiePose [13] samples poses with a score-based diffusion process on SE(3) and reaches high accuracy on T-LESS without symmetry annotations.

A diffusion denoiser, however, is an open-loop function. At each step it maps image features and the current noisy pose to a correction, and it never checks whether the current hypothesis is consistent with the image. The check is

the core of render-and-compare refinement [17, 18, 21, 42]: render the object at the hypothesis, compare the rendering with the observation, update. Render-and-compare has been the most accurate final stage in BOP for years [12, 38], but it requires a renderer and a textured 3D model, it runs as a separate refiner after an initial estimate, and it does not by itself represent a multi-modal distribution.

We ask whether the comparison can be moved into the diffusion process, in a learned feature space, so that every denoising step of every sample is conditioned on how well its hypothesis explains the image. This needs a model that answers the question "what would the image features be if the object were at pose g?" for arbitrary g, cheaply enough to be queried at every step. We build such a model as a world model in the sense of predictive latent models [3, 4, 45]: it operates on frozen DINOv2 [28] patch tokens, and predicts the tokens at a target pose from the tokens of a reference view and the relative pose between them. The reference is the nearest training view to the hypothesis, retrieved by pose, so the world model interpolates between nearby views instead of imagining the object from scratch. It is trained once per dataset with a pose-contrastive objective and then frozen.

Our method, LaRC (Latent Render-and-Compare), is a new architecture for diffusion-based pose estimation built around this world model (Fig. 1). It replaces the usual image-encoder-plus-pose-MLP denoiser with a closed loop: at every step, each sample retrieves a reference at its current pose, the world model predicts the tokens at that pose in the observation's crop, and a token transformer reads the observed tokens, the predicted tokens and their difference, conditioned on the noise level and the pose. The image encoder is a frozen foundation model, and the pose is grounded through a bank of posed training views rather than learned into network weights. The same predicted tokens define an energy, the foreground-weighted distance to the observation, which ranks a set of samples at test time without ground truth. We compare against a control that is identical to LaRC except that the predicted tokens are set to zero, so that the contribution of the comparison is measured in isolation.

We evaluate on four benchmarks with different sources of ambiguity: T-LESS [9] and YCB-V [43] under the BOP protocol [10], the SPEED+ spacecraft benchmark [31] with the official SPEC2021 score, and SYMSOL-T [13]. All our models, the control and LiePose are trained with the same number of training images. **[TODO: summarise the main findings in one paragraph once Tabs. 1–3 are filled; no claim before the numbers exist.]**

Our contributions are:

- A feature-space world model for object pose: it predicts DINOv2 patch tokens at an arbitrary pose from a retrieved reference view, with optional geometry tokens that make small pose changes explicit as image displacements (Sec. 4.2).
- LaRC, a closed-loop denoising architecture for SE(3) diffusion: a frozen foundation-model encoder, retrieval of posed reference views, and a token transformer that compares the world model's prediction at its current hypothesis with the observation at every step, together with an energy for selecting among samples without ground truth (Secs. 4.4 and 4.5).
- A controlled evaluation on four benchmarks against LiePose at a matched training budget and a no-world-model control, with analyses of when the comparison helps during sampling and what the world model can and cannot discriminate (Sec. 5).

2. Related Work

Instance-level 6D pose estimation. Methods for known objects either regress the pose directly [24, 41, 43], or predict 2D–3D correspondences and solve PnP [6, 8, 11, 22, 36], or compare the image with templates or codebooks of rendered views [26, 27, 37]. Symmetric objects make the image-to-pose map one-to-many; most methods handle this with symmetry-aware losses that need symmetry annotations [24, 41]. Our world model uses training views as references, as template methods do, but predicts features at a continuous pose instead of matching against a fixed set of templates.

Render-and-compare refinement. DeepIM [21] iteratively predicts a pose update from an observed and a rendered image; CosyPose [17], MegaPose [18] and FoundationPose [42] scale this to many objects and to novel objects. RePOSE [14] renders learned deep textures instead of RGB, and PixLoc [34] aligns learned feature maps for camera localisation. These methods compare in image or feature space, but they all need a renderer and a 3D model at test time and act as a refinement stage on a given initial pose. LaRC needs neither: the "rendering" is a prediction of frozen DINOv2 tokens from a real training view, and the comparison happens inside every step of a generative sampler, for every sample.

Diffusion and distributions on rotations and poses. Distributions over rotations have been modelled with implicit densities [25], normalising flows and mixtures; diffusion models on SO(3) [19] and SE(3) [40, 44] were introduced for protein structure, grasping and alignment. LiePose [13] applies score-based diffusion on SE(3) to object pose from images, with a surrogate Stein score and a ResNet image encoder. LaRC uses a different denoiser: a frozen encoder, retrieved reference views and a world model, so that in addition to observation features the network sees the predicted appearance at its own current hypothesis. Classifier guidance [7] steers a sampler with the gradient of an external

model; our world model enters as an input to the denoiser instead, and is used as an energy only to rank finished samples.

World models on pretrained features. Joint-embedding predictive architectures learn by predicting representations rather than pixels [3, 4]. DINO-WM [45] trains an action-conditioned predictor on frozen DINOv2 patch features and plans by optimising actions so that predicted features match a goal. We use the same principle with the object pose as the "action": the predictor maps a reference view's tokens and a relative pose to the tokens at the target pose, and the denoiser uses the prediction to reason about its hypothesis.

Spacecraft pose estimation. SPEED [15] and SPEED+ [31] benchmark the pose of a known spacecraft, with SPEED+ testing on hardware-in-the-loop images (lightbox, sunlamp) after training on synthetic renders only. Leading methods regress keypoint heatmaps and solve PnP [5], often with domain randomisation, style augmentation or adaptation on the unlabeled test domains [29, 31, 32]. The spacecraft has near-symmetric configurations that produce flipped estimates, which makes it a natural test of a sampler that can represent and resolve ambiguity.

3. Background: The Generative Process

Pose parametrisation. Following LiePose [13] and GDR-Net [41], a pose is expressed relative to an image crop I of the object: the rotation is allocentric [16], *i.e.* it removes the rotation induced by the object's position in the image, and the translation is scale-invariant [22], $t = (\delta_x/w,\ \delta_y/h,\ z/r)$, with (δ_x, δ_y) the offset of the projected object centre from the crop centre, (w, h) the box size and r the crop's zoom ratio. We write $g = (R, t) \in \mathrm{SE}(3)$ for this crop-relative pose; it maps to the camera frame given only the crop box.

Diffusion on SE(3). The forward process [13] perturbs a clean pose g_0 by Gaussian noise in the Lie algebra, $g_t = g_0 \,\mathrm{Exp}(\sigma_t \epsilon)$, $\epsilon \sim \mathcal{N}(0, I_6)$, with a variance-exploding power-law schedule σ_t over $T = 100$ steps. A network $\epsilon_\theta(g_t, t, I)$ is trained to predict the noise, and sampling starts from a uniform pose and applies T DDIM-style [35] updates on the group. Each training image is perturbed into many noisy poses ("slices") per step. In LiePose, ϵ_θ is a ResNet-34 image encoder followed by Fourier-feature MLP blocks that take the noisy pose and the time step.

What the denoiser does not see. ϵ_θ receives the image once, as a feature vector, and the hypothesis g_t as a vector. Whether the object would look like I at g_t is never computed explicitly; the network must learn it implicitly for every pose in SE(3). Section 4 introduces a denoiser architecture that computes it: an explicit prediction of the appearance at g_t, compared with the observation.

FIGURE PLACEHOLDER (architecture).
(a) World model: reference tokens $\mathbf{z}^{\mathrm{ref}}$ (+ geometry tokens, + 7 virtual-point tokens) $\rightarrow$ 6 adaLN transformer blocks conditioned on Fourier features of $(g,\ g\,(g^{\mathrm{ref}})^{-1})$ $\rightarrow$ predicted tokens $\hat{\mathbf{z}}(g)$.
(b) Denoiser step: $g_t \rightarrow$ retrieval $\rightarrow \mathcal{F}_\phi$ (frozen) $\rightarrow [\mathbf{z}, \hat{\mathbf{z}}, \hat{\mathbf{z}} - \mathbf{z}, w]$ tokens + conditioning token $c(t, g_t, k) \rightarrow 4$ adaLN blocks $\rightarrow \epsilon_\theta$.
(c) Selection: N samples, energy $E(g_i)$, arg min.

Figure 2. **Architecture of LaRC.** (a) The world model predicts tokens at pose g from the nearest training view. (b) Each denoising step compares this prediction at the current hypothesis with the observation. (c) The same prediction gives an energy that ranks finished samples.

4. Method

LaRC has two parts. A world model $\mathcal{F}_\phi$ predicts the DINOv2 tokens of an object at an arbitrary pose from a retrieved reference view (Secs. 4.1 and 4.2); it is trained first and then frozen (Sec. 4.3). A diffusion denoiser uses the prediction at its current hypothesis at every step (Sec. 4.4), and the same prediction ranks complete samples (Sec. 4.5). Figure 2 gives an overview.

4.1. Observation tokens

The input is a 224×224 crop I of the object, cut and resized as in LiePose (dynamic zoom-in at training time), with pixels outside the visible mask set to zero when a mask is part of the protocol (Sec. 4.6). A frozen DINOv2 ViT-S/14 [28] maps it to 16×16 patch tokens of dimension 384, which we average-pool 2×2 into $\mathbf{z} \in \mathbb{R}^{64 \times 384}$. A per-token weight $w_p = w_0 + m_p$ emphasises the object: $m_p \in \{0, 1\}$ is the max-pooled visible mask dilated by one token, and $w_0 = 0.05$ keeps a small weight on background tokens. Where no mask is available (SPEED+), $m_p = 1$. The crop is also described by box quantities k (zoom, crop origin, box size and aspect), which contain no pose information. Tokens of all training crops are computed once and cached; the image encoder is never trained.

4.2. A retrieval-referenced world model

Predicting the appearance of an object at an arbitrary pose from nothing is hard: the model would have to store the ob-

ject's appearance from every viewpoint. We instead predict from the nearest available view. For each object o, the *reference bank* holds the cached tokens, poses and crop boxes of the training crops, $\mathcal{B}_o = \{(\mathbf{z}_i, g_i, k_i)\}$, expanded by the object's symmetry group $\mathcal{S}_o$ when one is known: each crop appears once per symmetric equivalent $(R_i S, t_i)$, $S \in \mathcal{S}_o$, subsampled to at most 36 (T-LESS, YCB-V) or 60 (SYMSOL-T) equivalents. Given a query pose $g = (R, t)$, retrieval returns the nearest entry under the key $[\mathrm{vec}(R_{\mathrm{allo}}),\ \lambda t]$, $\mathrm{ret}(g) = (\mathbf{z}^{\mathrm{ref}}, g^{\mathrm{ref}}, k^{\mathrm{ref}})$. While training both the world model and the denoiser, entries that come from the observation's own source image are excluded, so the model never sees its target as a reference. At test time the bank contains training crops only.

The world model predicts the tokens of the object at pose g, in the crop k of the observation:

$$\hat{\mathbf{z}}(g) = \mathcal{F}_\phi\big(\mathbf{z}^{\mathrm{ref}},\ g^{\mathrm{ref}} \to g,\ k^{\mathrm{ref}},\ k\big) \in \mathbb{R}^{64\times 384}. \quad (1)$$

$\mathcal{F}_\phi$ is a 6-block transformer of width 384 whose blocks are modulated (adaLN [33]) by Fourier features of the target pose g and of the relative pose $g\,(g^{\mathrm{ref}})^{-1}$. Its input tokens are a linear projection of $\mathbf{z}^{\mathrm{ref}}$.

Geometry tokens. A rotation of half a degree changes DINOv2 tokens very little, but it moves image points by a computable amount. When the camera intrinsics and the crop box are available, we give this displacement to the model explicitly. For every reference token centre we back-project onto a fronto-parallel plane at the reference object depth, move the point rigidly from g^{ref} to g, and project it into the observation's crop; the token receives the displacement in token units and the log depth ratio, Fourier-encoded and added through a zero-initialised layer. In addition, seven virtual object-frame points (the centre and six axis tips at a nominal radius; no CAD model) are projected at both poses and prepended as tokens. We use geometry tokens on SYMSOL-T and SPEED+; on T-LESS and YCB-V the token cache does not store the per-crop intrinsics they need, and the world model uses the reference tokens and pose only. **[TODO: state whether YCB-V uses geometry tokens once its world model is trained.]**

Energy. The world model defines an energy of a pose hypothesis given the observation, the foreground-weighted squared token distance

$$E(g; \mathbf{z}) = \frac{\sum_p w_p \,\frac{1}{D}\|\hat{\mathbf{z}}_p(g) - \mathbf{z}_p\|^2}{\sum_p w_p}, \quad (2)$$

with $D = 384$ and the reference retrieved at g.

4.3. Training the world model

We train $\mathcal{F}_\phi$ so that its energy ranks poses, not only so that it reconstructs tokens. For each training crop with pose $g^\star$ we form a candidate set $\mathcal{C}$: the ground truth, 11 uniformly random rotations, 12 rotations within 30° of $g^\star$, 4 translation perturbations and 4 translations taken from other crops. The reference is retrieved at a noisy copy of $g^\star$ (rotation perturbed by $|\mathcal{N}(0, 5^\circ)|$, translation by Gaussian noise), which mimics the poses at which the denoiser will query it. With a soft target that accounts for symmetry, $q_c \propto \exp\!\big(-d_{\mathcal{S}}(R_c, R^\star)^2/2\sigma_R^2 - \|t_c - t^\star\|^2/2\sigma_t^2\big)$, where $d_{\mathcal{S}}$ is the rotation distance to the nearest symmetric equivalent, the loss is

$$\mathcal{L}_{\mathrm{WM}} = \mathrm{CE}\Big(\mathrm{softmax}_c\big(-E(g_c)/\tau\big),\ q\Big) + \lambda_{\mathrm{rec}}\, E(g^\star), \quad (3)$$

with a learned temperature τ. The first term makes the energy rank candidates; the second anchors the prediction at the true pose to the observed tokens. On SYMSOL-T we add regression targets from in-plane rotations of training images and from rendered pose pairs (Sec. B of the supplement). The world model is trained once per dataset and is not updated afterwards.

4.4. Comparing inside every denoising step

At step t, each sample with pose g_t retrieves $\mathrm{ret}(g_t)$ and receives the world model's prediction $\hat{\mathbf{z}}(g_t)$ in its observation's crop. The denoiser forms one input token per image patch from the observation, the prediction, their difference and the weight,

$$x_p = \mathrm{Lin}\big(\mathrm{LN}([\,\mathbf{z}_p,\ \hat{\mathbf{z}}_p(g_t),\ \hat{\mathbf{z}}_p(g_t) - \mathbf{z}_p,\ w_p\,])\big) + e_p, \quad (4)$$

with a learned position embedding e_p. A conditioning vector $c = \mathrm{Lin}(\mathrm{SiLU}(\gamma_t(t) + \gamma_g(r_t) + \gamma_k(k)))$ combines Fourier embeddings γ of the step, of the Lie-algebra coordinates r_t of g_t, and of the crop box. c is prepended as a token and modulates four adaLN transformer blocks of width 256; the output at the conditioning token is mapped to the 6-dimensional noise ϵ_θ. The prediction $\hat{\mathbf{z}}$ is treated as a constant (no gradient reaches $\mathcal{F}_\phi$).

The network is trained and sampled with the generative process of Sec. 3 (noise-prediction loss, 100-step sampler, weight averaging). Because $\hat{\mathbf{z}}$ depends on g_t, the denoiser can read the error of its own hypothesis from the difference tokens instead of inferring it from image features alone. Retrieval is a nearest-neighbour query on the CPU and the world model is a forward pass per sample and step; we report the cost in Sec. 5.5.

Control without the world model. To isolate the comparison, the control is the same network with $\hat{\mathbf{z}} := 0$ in Eq. (4), trained identically. It sees the same DINOv2 observation tokens, crop box and pose embedding, so any difference between LaRC and the control is due to the predicted tokens alone. Note that the control is a DINOv2 token denoiser, not LiePose, which uses a trained ResNet-34 encoder.

4.5. Selecting among samples

A diffusion sampler returns one pose per run, and independent runs can land in different modes, *e.g.* flipped configurations of a near-symmetric object. We draw N samples from independent priors and return $\hat{g} = \arg\min_i E(g_i; \mathbf{z})$ (Eq. (2)), with the reference retrieved at each g_i. Selection uses only the observation and the training bank. N is fixed in advance for all benchmarks, $N = 16$; results with one sample are the protocol of the main tables unless marked otherwise.

4.6. Information used at test time

At test time LaRC uses the image crop, the box quantities k, the camera intrinsics (for geometry tokens), the object identity (to select the reference bank $\mathcal{B}_o$, as in the BOP task where the target objects are given), and the cached tokens of training crops. Crops are ground-truth boxes on all benchmarks: on T-LESS and YCB-V the visible box and visible mask, as in LiePose's protocol [13]; on SPEED+ the box of the spacecraft keypoints projected with the ground-truth pose; on SYMSOL-T the box of the rendered object's foreground. On T-LESS, YCB-V and SPEED+ every method we train uses the same crops. On SYMSOL-T LiePose takes the full image, so the crop gives LaRC information that LiePose does not use; we report LiePose with the same crops as well (Tab. 3). Rows whose crops come from a detector are marked. Apart from these crops, no ground-truth quantity enters inference.

5. Experiments

5.1. Setup

Benchmarks. *T-LESS* [9]: 30 texture-less industrial objects with discrete and continuous symmetries, heavy clutter and occlusion. We train on the PBR and real training images and test on the BOP19 targets of the real test set. *YCB-V* [43]: 21 household objects in cluttered video frames; PBR and real training images, BOP19 test targets. *SPEED+* [31]: a single spacecraft; we train only on the synthetic images and test on the full hardware-in-the-loop *lightbox* and *sunlamp* sets, without using their images in any form during training. *SYMSOL-T* [13]: five symmetric shapes (tetrahedron, cube, icosahedron, cone, cylinder) with random rotations and translations; LiePose's training and test splits.

Metrics. On T-LESS and YCB-V we report the BOP average recalls of VSD, MSSD and MSPD and their mean AR [12], computed with the BOP toolkit; VSD uses the toolkit's Python renderer (Sec. D). On SPEED+ we report the mean translation error e_t (m), the mean rotation error $e_q = 2\arccos|\langle q, \hat{q}\rangle|$ (deg), and the official SPEC2021 score $S = \frac{1}{n}\sum(e_q^{\text{rad}} + e_t/\|t\|)$ with the competition's precision thresholds [31] (Sec. D), on the full test sets. On SYMSOL-T we report the mean symmetric rotation error (deg) and mean translation error per shape, as in LiePose.

Baselines and training budget. The main baseline is LiePose [13], which we retrain on every benchmark with its public code. All models we train see the same number of training images as our LiePose runs: 400k steps with batch 64 on T-LESS, YCB-V and SPEED+, and 400k steps with batch 32 on SYMSOL-T (LiePose: 800k steps with batch 16). Our T-LESS LiePose run uses batch 64 where the paper uses 32, and the control and LaRC draw from a fixed set of pre-augmented crops (Sec. F). Because every noisy pose of LaRC queries the world model, we use 64 noisy poses per image instead of LiePose's 256; the control uses the same 64. We evaluate the final checkpoint of every run; no checkpoint is chosen on test data. For each run we sample with the weight average (EMA) or the live weights, whichever has the lower training loss, decided before any test evaluation (Sec. E). Published numbers of other methods are copied from their papers and marked by protocol.

Implementation. The world model has 6 blocks of width 384 and is trained for 30k steps (20k on SYMSOL-T) with AdamW. The denoiser has 4 blocks of width 256. Sampling uses LiePose's 100-step sampler. Full details are in Sec. A.

5.2. Main results

T-LESS and YCB-V. Table 1 compares LaRC with LiePose trained by us under identical crops, data and training budget, with the no-world-model control, and with published methods. **[TODO: describe: (i) LaRC vs control (the effect of the comparison), (ii) LaRC vs LiePose (ours), (iii) where the difference comes from, using the rotation and translation accuracies of Tab. 3, (iv) the effect of selection. Do not compare GT-crop rows with detector rows.]**

SPEED+. Table 2 reports the full lightbox and sunlamp sets. **[TODO: describe LaRC vs control vs LiePose; the effect of geometry tokens; the effect of selection, and the gap to the oracle as an upper bound on what selection could reach. State explicitly that the SPEC2021 entries used the unlabeled test-domain images and are not under our protocol.]**

SYMSOL-T. Table 3 reports per-shape errors. **[TODO: describe; compare LiePose with crops vs with the full image first, so that the effect of the crop is separated from the effect of the world model.]**

5.3. Ablations

What the world model contributes. Table 4 replaces the world model while keeping the denoiser and budget fixed.

Table 1. **T-LESS and YCB-V** (BOP19 targets, BOP average recall, %). *Crops*: GT = ground-truth visible box and mask; Det = detector boxes. Rows with different crops are not directly comparable. *Ours* = trained by us with the same data, crops and number of training images. Published numbers are rounded to one decimal. †AR not reported; mean of the three recalls, computed by us. The GDRNPP numbers that [13] compares with are GDRNPP's leaderboard entry with one model per dataset and detected boxes (67.6 / 75.1 / 90.2), not a ground-truth-crop result. Published rows train on PBR and real images; §per-metric recalls from the BOP leaderboard [1], as the papers report AR only (GDRNPP) or no BOP numbers (ZebraPose). **[TODO: YCB-V of our rows.]**

				T-LESS				YCB-V			
Method	Encoder	Crops	Samples	VSD	MSSD	MSPD	AR	VSD	MSSD	MSPD	AR
Published, detector crops (BOP leaderboard, RGB only)											
CosyPose [17]	–	Det	1	66.9	69.5	82.1	72.8	77.2	84.2	85.0	82.1
SurfEmb [8]	–	Det	1	–	–	–	77.0	–	–	–	71.8
ZebraPose [36]§	–	Det	1	69.6	74.0	88.9	77.5	75.1	83.0	86.4	81.5
CIR, RGB [23]	–	Det	1	66.3	68.4	79.8	71.5	78.3	83.5	85.2	82.4
GDRNPP [24]§	–	Det	1	68.7	76.0	91.3	78.6	76.0	84.6	86.9	82.5
Published, ground-truth crops											
LiePose [13]	ResNet-34	GT	1	56.9	60.2	93.2	70.1†	–	–	–	–
Trained by us, ground-truth crops, matched number of training images											
LiePose (our training)	ResNet-34	GT	1	63.1	65.7	92.3	73.7				
Control (no world model)	DINOv2-S	GT	1	30.5	34.0	78.3	47.6				
LaRC	DINOv2-S	GT	1	38.4	42.3	82.4	54.3				
LaRC + selection	DINOv2-S	GT	N								

Table 2. **SPEED+** (full lightbox and sunlamp sets; e_t in m, e_q in degrees, S = official SPEC2021 score; lower is better). All methods train on synthetic images only, except where marked. *Box*: GT = box of the keypoints projected with the ground-truth pose; Det = detected. The oracle row selects with the ground truth and is a diagnostic, not a result. S applies the precision floors to rotation and translation separately, as the competition server does; *floors applied jointly as in [31] (at most 0.005 lower per image). –: not reported in metres. Published e_q rounded to 0.1°. EagerNet uses test-time augmentation and several hypotheses. ¶Best submission of each team per category on the final leaderboard [2], e_q as tabulated in [30]. **[TODO: fill our rows.]**

			lightbox			sunlamp		
Method	Box	Samples	e_t	e_q	S	e_t	e_q	S
Published, synthetic training only								
KRN [31]*	GT	1	2.25	44.5	1.12	14.64	81.0	3.73
HigherHRNet + EPnP [31]*	–	1	0.97	34.7	0.77	0.85	47.8	0.98
SPNv2 (ϕ=3, BN) [29]	–	1	0.175	6.5	0.142	0.225	11.1	0.230
EagerNet [39]	Det	–	–	1.7	0.039	–	2.7	0.059
SPNv3-B [30]	GT	1	–	2.2	0.051	–	3.6	0.079
Legrand et al. (HRNet) [20]	GT	1	0.09	4.3	0.09	0.14	6.9	0.14
Published, using unlabeled lightbox/sunlamp images (different protocol)								
SPNv2 + online domain refinement [29]	–	1	0.142	5.6	0.122	0.182	9.6	0.198
SPEC2021, TangoUnchained [32]¶	–	1	–	3.2	0.073	–	4.3	0.090
SPEC2021, lava1302 [32]¶	–	1	–	6.7	0.165	–	2.7	0.059
Trained by us, synthetic only, GT keypoint box, matched number of training images								
LiePose (our training)	GT	1	0.183	11.6	0.234	0.175	14.9	0.289
Control (no world model)	GT	1						
LaRC, retrieval world model	GT	1						
LaRC, + geometry tokens	GT	1	0.317	37.0	0.701	0.275	38.3	0.716
LaRC, + geometry tokens + selection	GT	N						
Oracle selection among the same N (diagnostic)	GT	N						

Table 3. **SYMSOL-T** (mean rotation error R in degrees, mean translation error t; lower is better). *Input*: full image or crop around the rendered object. **[TODO: fill.]**

		tet.		cube		icosa.		cone		cyl.	
Method	Input	R	t	R	t	R	t	R	t	R	t
Regression [13]	full	2.92	0.064	2.86	0.050	2.46	0.037	1.84	0.058	2.24	0.049
Iter. regression [13]	full	4.25	0.048	4.20	0.037	29.33	0.026	1.63	0.037	2.34	0.032
LiePose [13]	full	0.59	0.016	0.58	0.011	0.64	0.012	0.54	0.016	0.41	0.011
LiePose (ours)	full										
LiePose (ours)	crop										
Control	crop										
LaRC	crop										

Table 4. **World-model variants.** Same denoiser and training budget (100k steps × batch 32, 16 noisy poses per image; Tab. 5); one sample. T-LESS: AR (%, ↑); SPEED+: score S on lightbox (↓); SYMSOL-T: mean rotation error over shapes (deg, ↓). **[TODO: fill.]**

World model in the denoiser	T-LESS	SPEED+	SYMSOL-T
None (control, $\hat{\mathbf{z}}$=0)	41.2	0.782	
Learned template, no reference			
Retrieved reference	54.7	0.734	
+ geometry tokens	–	0.717	
+ roll / render-pair targets	–	–	
Oracle $\hat{\mathbf{z}}$: GT-view tokens (diag.)			

Table 5. **Denoiser inputs and when the comparison is used.** T-LESS AR (%, ↑) and SPEED+ lightbox S (↓), one sample, 100k training steps × batch 32 (16 noisy poses per image) for every row. **[TODO: fill.]**

Variant	T-LESS	SPEED+
$[\mathbf{z}]$ (control)	41.2	0.782
$[\mathbf{z}, \hat{\mathbf{z}}]$	53.7	0.716
$[\mathbf{z}, \hat{\mathbf{z}}, \hat{\mathbf{z}} - \mathbf{z}]$	54.6	0.718
$[\mathbf{z}, \hat{\mathbf{z}}, \hat{\mathbf{z}} - \mathbf{z}, w]$ (LaRC)	54.7	0.734
Comparison only for $t \geq t_0$ (high noise)	40.7	0.774
Comparison only for $t < t_0$ (low noise)	52.4	0.755
Energy only (control + selection)		
8×8 pooled tokens (LaRC)	54.7	0.734
16×16 full-resolution tokens		

Table 6. **Selection among N samples** (SPEED+ lightbox S, ↓; T-LESS AR, ↑). Energy: Eq. (2); random: a uniformly random sample; oracle: the sample closest to the ground truth in rotation (diagnostic). SPEED+: the 100k model with geometry tokens (Tab. 4). **[TODO: fill.]**

	SPEED+ lightbox S			T-LESS AR		
N	random	energy	oracle	random	energy	oracle
1	0.713	0.713	0.713			
4						
8						
16	0.713	0.593	0.331			

Table 7. **What the world model can discriminate.** For held-out observations, the probability that the energy of a perturbed pose exceeds the energy of the ground-truth pose (0.5 = chance), by size of the rotation perturbation, and for translation perturbations; references are retrieved at a noisy ground truth. **[TODO: fill; T-LESS from held-out training crops, SPEED+ from the validation split, SYMSOL-T after the evaluation set is rebuilt from training images (Sec. F).]**

	rotation perturbation			
	1–3°	3–10°	10–30°	translation
T-LESS				
YCB-V				
SPEED+ (retrieval)				
SPEED+ (+ geometry)				
SYMSOL-T				

A per-object learned template predicts tokens from the pose alone, without a reference view; the retrieval variants predict from the nearest training view. The oracle row feeds the tokens of the ground-truth view and is an upper bound on what a perfect world model could provide. **[TODO: describe.]**

How the denoiser uses it. Table 5 varies the denoiser's input tokens, restricts the comparison to part of the noise schedule (the prediction is zeroed, as in the control, outside $t \geq t_0$ or $t < t_0$, with $t_0 = 50$ of the 100 training timesteps, fixed in advance), and compares in-loop comparison against using the world model only to select among samples of the control. The last block varies the token resolution of both observation and prediction. **[TODO: describe.]**

Selection. Table 6 varies the number of samples. The oracle column shows how often a good sample exists among the N, and therefore how much any selection rule could gain; the energy column shows how much of this the world model recovers. **[TODO: describe.]**

FIGURE PLACEHOLDER (flip timing, SPEED+ lightbox, diagnostic).
x: denoising step (0 = pure noise, 100 = final). Left y: share of samples whose rotation error exceeds 90°. Right y: within-image probability that the energy prefers a good sample ($< 30°$) over a flipped one ($> 90°$). Curves for LaRC and control.

Figure 3. **When are flips decided?** The share of flipped samples and the world model's ability to tell flipped from correct hypotheses, along the sampling trajectory. Ground truth is used only to measure. **[TODO: caption text from the result.]**

FIGURE PLACEHOLDER (qualitative).
Rows: T-LESS, YCB-V, SPEED+ lightbox, SPEED+ sunlamp. Columns: input crop; LiePose estimate (projected model outline); LaRC estimate; predicted tokens at the estimate vs observed tokens (PCA colours); one failure case per dataset.

Figure 4. **Qualitative results and failures.** **[TODO: caption.]**

5.4. Analysis

What the world model can and cannot see. Table 7 measures the world model on its own, independently of the denoiser: how reliably its energy prefers the true pose over perturbed ones. This bounds the precision that the comparison can add on top of the denoiser. **[TODO: describe; relate the finest bin to the remaining error of LiePose and LaRC in Tab. 1.]**

When are flips decided? On SPEED+, the dominant failure is a flipped estimate. Figure 3 tracks, along the sampling trajectory, the share of samples that are flipped and how well the energy separates flipped from correct samples of the same image. **[TODO: describe; this analysis motivates selection (Sec. 4.5) and the noise-level ablation of Tab. 5.]**

5.5. Runtime

Table 8 breaks down the inference cost. The world model and the retrieval are evaluated once per sample and step. **[TODO: describe; mention fewer steps as in LiePose's runtime study if measured.]**

Table 8. **Runtime** per image (ms), one sample, 100 steps, batch 100, on one **[TODO: GPU model]**. Retrieval runs on the CPU. **[TODO: fill.]**

Method	encoder	retrieval	world model	total
LiePose (ours)		–	–	
Control		–	–	
LaRC				
LaRC + selection (N)				

6. Limitations and Conclusion

Limitations. LaRC relies on ground-truth crops on all four benchmarks, as LiePose does on T-LESS; with detected boxes and masks the inputs of the world model and of the denoiser change, and we have not evaluated this setting. The world model needs a bank of posed training views of each object and the object's identity at test time; it does not apply to unseen objects. Its precision is bounded by what frozen, pooled DINOv2 tokens can distinguish (Tab. 7), and the retrieval and world-model queries at every step make sampling slower than LiePose (Tab. 8). Geometry tokens require the camera intrinsics and the crop box. Finally, our runs use 64 noisy poses per training image instead of LiePose's 256. **[TODO: revise once the results are in; add any failure mode seen in Fig. 4.]**

Conclusion. We moved render-and-compare into SE(3) diffusion: a frozen world model predicts, at every denoising step and for every sample, the DINOv2 tokens of the object at the current hypothesis, and the denoiser conditions on their comparison with the observation. The same prediction gives an energy that ranks samples without ground truth. Because the diffusion formulation is unchanged and the control differs only by the predicted tokens, the experiments isolate what the comparison contributes. **[TODO: one or two sentences of findings, written after the tables are filled.]**

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